THE FLEET — ONE NETWORK, MANY SHAPES

NODE	NAME	ROLE	ENVELOPE	POWER	FINISH	STATUS
L1	Brick	field node — mailbox+repeater+bulletin+SOS	160×80×24 mm	4×18650/solar	graphite PETG matte	CAD ready
E	Totem	community board — public E-Ink + mesh node	~320×220×80+pole	solar+LiFePO4	charcoal PC IP66	spec
S	Mite	swarm unit — perchable env sensor	~90×90×35 mm	solar+LiPo	forest ASA matte	CAD scripted
A	Anchor	charging station — dock+relay+backhaul	~350×250×400+array	20–60 W+LiFePO4	charcoal powder-coat	spec
P	ArcticPhone	rural comms — mesh-native, sat SOS	~145×72×14 mm	replaceable+solar	graphite PC+TPU	concept
D	Seed	degradable mote — seasonal sensing	~20×20×5 mm	printed+harvest	cellulose — vanishes	R&D
H	Meridian	satellite payload — mesh IS ground segment	1U/3U CubeSat	solar array	aluminum 1U	vision

SHARED DESIGN DNA (EVERY NODE)

— Climate: -45..+70 °C · IP65+ · UV-stable · LiFePO4-preferred · December-sized solar.
— Power: solar-first · 30–90 day autonomy · works with zero grid.
— Field: bear-country mounts · two-person install · hand-tool repair.
— Sovereignty: community-owned mesh · data stays home · OCAP · EN/FR · privacy by design.
— Appropriate tech: right-sized cost · repairable > sealed · open docs.
— Offline dignity: works when everything else has failed — that IS the product.

DRAWING INDEX

AF-000 this sheet — fleet overview & index
AF-L1-101…105 L1 Brick — arrangement, parts/power, assembly, electronics, finish
AF-S-101…107 S Mite — arrangement, parts, assembly, electronics, variants, mechanical, finish
AF-E-101…103 E Totem — arrangement, parts/install, finish
AF-A-101…103 A Anchor — arrangement, parts/power, finish
AF-P-101…107 P ArcticPhone — concept, spec, definition, tri-radio, modules, UX, finish
AF-D-101…103 D Seed — R&D brief, materials/degradation, cluster protocol
AF-H-101 H Meridian — satellite mission concept
AF-X-101…102 Component matrices — radios, compute, power, displays, enclosures

HOW TO READ THIS SET

— L1 Brick = build-ready (BOM, wiring, electronics, assembly, acceptance test).
— S Mite = prototype-ready (orderable parts, 4 swarm variants, CAD scripted).
— Totem/Anchor = spec-level · Phone = concept · Seed/Meridian = R&D/vision.
— Every product has a FINISH & MATERIALS sheet — final look + materials spec.
— AF-X matrices = every electronic decision — radios, compute, power, displays.
— Every sheet: SURAFEL NEGUSSIE drew it · EMEBET INC. owns it · PROPRIETARY.

ARCTICCHAIN — RURAL NODE FLEET
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